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CLINICAL FEATURE
REVIEW

Role of surgical thoracic sympathetic interruption in treatment of facial blushing: a systematic review

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ABSTRACT

Objectives: This paper aims to review the evidence to support the effectiveness of sympathectomy as a treatment for facial blushing in terms of relief of facial blushing, patient satisfaction, recurrence of blushing, patients regretting treatment and its associated complications.

Methods: A systematic search strategy was performed in Ovid-Medline, Embase, Cochrane library and NICE. Studies reporting outcomes of sympathetic interruption in the treatment of facial blushing were retrieved.

Results: Nine studies met the inclusion criteria with 1369 patients included in the final analysis. The age range of patients was 8 to 74 years (from 7 studies) with 56% females. Mean follow up was 21 months in 8 studies (range 6 to 30 months). The pooled proportion of patients who had good relief of facial blushing was 78.30% (95% C.I. 58.20% – 98.39%). Complete satisfaction was reported in 84.02% (95% C.I. 71.71% – 96.33%). Compensatory sweating and gustatory sweating were the commonest complications occurring in 74.18% (95% C.I. 58.10% – 90.26%) and 24.42% (95% C.I. 12.22% – 36.61%) respectively. The estimated proportion of patients regretting surgery was 6.79% (C.I 2.08% 11.50%).

Conclusion: Sympathetic interruption at T2 or T2-3 ganglia appears to be an effective treatment for facial blushing. However, lack of randomized trials comparing sympathetic interruption with non-surgical methods of treatment and heterogeneity of included studies with respect to assessment of outcome measures preclude strong evidence and definitive recommendations.

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KEYWORDS
Blushing; flushing; sympathectomy; autonomic nerve block

1. Introduction

Facial blushing (syn. erythrophobia) is an involuntary condition caused by autonomic dysfunction [1]. It presents as disproportionate redness of the face, ears, neck, and upper chest in response to heightened anxiety states. It usually occurs in young individuals and is a significant problem in the fair skinned. The exact prevalence of facial blushing is not known, but, it forms one of the prominent conditions in social phobia, the prevalence of which is around 10% [2,3].

Various treatment strategies have been used to manage facial blushing. Social anxiety disorders are identified and treatment targeted toward them using various forms of psychotherapy before offering treatment for blushing per se [4–10]. Selective serotonin reuptake inhibitors (SSRI), like sertraline[4] and beta blockers, such as atenolol[11], can be useful adjuncts in the treatment of facial blushing before considering surgery. Surgical treatment is usually reserved for patients who are severely affected in whom anxiety disorders are ruled out. Thoracoscopic sympathectomy is the most common choice of intervention, although percutaneous techniques employing radiofrequency and chemical ablation of the sympathetic ganglia have also been described. There are no focused systematic reviews addressing role of sympathetic interruption in facial blushing alone [12]. This review considers the evidence in support of sympathetic interruption in the treatment of facial blushing, its outcomes and associated complications.

2. Objectives

To review the evidence to support effectiveness of sympathectomy or sympathetic blockade as treatment for facial blushing in terms of relief of blushing, patient satisfaction, recurrence of blushing, patients regretting treatment and its associated complications.

3. Methods

The methodology of the current review was according to the Preferred Reporting Items for Systematic Reviews and Meta-

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Analyses (PRISMA) guidelines [13]. The study is registered with Centre for Reviews and Dissemination, University of York, PROSPERO 2016:CRD42016033017.

3.1. Inclusion and exclusion criteria

Studies on patients with facial blushing, who underwent sympathetic interruption in which data for facial blushing could be separately extracted, were included. All types of study design such as randomized controlled trials, nonrandomized controlled trials, descriptive studies, and case series were included.

In this review, sympathetic interruption includes sympathetectomy and sympathetic clipping or blockade. As per the nomenclature for the level of sympathetic interruption, an expert consensus published in 2011 on management of hyperhidrosis advocates adopting rib rather than vertebrae as reference level [14].

Studies with patients who had previous sympathectomy or re-do procedures for facial blushing were excluded. This included secondary causes of blushing namely, neurogenic rosacea, Harlequin syndrome, Frey's syndrome, and menopause. Experimental animal studies and studies reporting insufficient preoperative and postoperative data were also excluded.

3.2. Search methods for identification of studies

The electronic databases searched to identify potential studies were Ovid-Medline, Embase, Cochrane library, and NICE. No limits or filters were applied to any of the databases as we found the literature on the topic of interest was limited. The key words used for the search were 'blushing,' 'flushing,' 'facial flush,' 'sympathectomy,' 'sympathetic clipping,' 'sympathetic block,' 'sympathectomy,' 'nerve block,' and 'nerve clipping.' However, a tailor made search strategy was created for each of the databases reviewed (Supplemental data).

3.3. Selection of studies

With the specified search strategy, each database was explored by two authors (GG and RD) independently. Title and abstract screening was done for all the articles obtained through the initial hit. The search results were merged to remove duplication. Articles were rejected on initial screen when the review authors determined from the title and the abstract that they were not relevant. When a title or abstract was inconclusive to include or reject, full text was examined. The two review authors independently assessed the studies for inclusion and resolved the differences by discussion and, when necessary, in consultation with a third review author (Baker D M). In the case of duplicate publications and companion papers of a primary study, all available data were obtained for complete information and the studies were linked together. In cases of doubt, the original publication (usually the oldest version) obtained priority and full text was examined for compliance of the study with eligibility criteria. When information regarding any of the above was unclear, an attempt was made to contact the authors. Figure 1 (PRISMA chart) illustrates the selection process of studies included in the review.

3.4. Data extraction and management

Data extraction form was constructed based on the review objectives. Two review authors independently extracted data from included papers on to an Excel spread sheet. Items under which data were extracted include – authors, institution, country, period of study, type of study, inclusion, exclusion, age (average, mean and range), total number of patients, gender distribution, duration of symptoms, range of duration of symptoms, initial assessment, medical treatment, procedure performed including number of cases, level of sympathetic interruption, post-op result assessment tool, post-op result (relief of facial blushing and patient satisfaction), complications, regret undergoing surgery, follow-up period, drop out, funding, conflict of interest, and ethical approval.

3.5. Assessment of methodological quality of included studies

Case series comprised the majority of studies in the systematic review [15–21]. In this review, the tool from National Heart, Lung and Blood Institute of NIH [22] was used to assess the quality of case series designs. Accordingly, in the final analysis, we included four studies which were considered 'good' [15–18] and three as 'fair' [19–21].

A randomized controlled trial by Licht et al. [23] was assessed for risk of bias using the criteria outlined in the Cochrane Handbook for Systematic Reviews of Interventions [24]. The study was found to have 'low risk of bias' in key domains except the absence of control group which was considered as 'high risk of bias' in the 'other bias' category.

The intervention trial by Jadresic et al. [4] was categorized as 'controlled before-and-after study.' For this particular study, the Cochrane Collaboration's tool for assessing risk of bias was adapted using EPOC's criteria for studies other than RCTs [25]. According to this tool, the study had 'low risk of bias' in three key domains and 'high risk of bias' in four key domains. However, for the purpose of our review, the data from this interventional trial is considered as similar to a case series and hence included in the review.

3.6. Data analysis

Proportion of each parameter and its confidence interval was computed. Heterogeneity was assessed using Q statistics and I^2. Meta-analysis was carried out using random effects model in STATA 13. We included all the nine studies if they reported specific outcome measures. We attempted subgroup analysis based on the study design, as we had seven case series and two intervention trials. However, considering the methodological limitations of the two intervention trials, we combined all the study designs and discussed the results in detail. This approach is justified as our main interest was on the outcome of sympathectomy and not in the comparison of levels of sympathectomy [23] or sympathectomy versus medical treatment [4]. The data of outcome measures related to sympathectomy were isolated from the two intervention trials [4,23] and were used for meta-analysis. Results are expressed in terms of proportion, 95% confidence interval and forest plot.

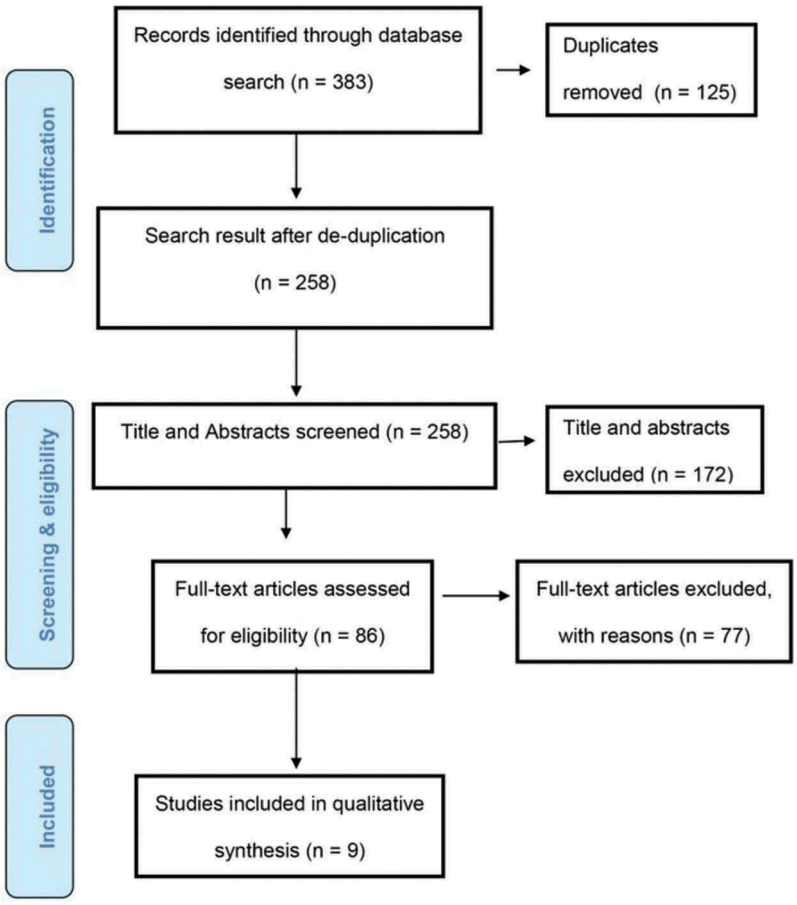

Figure 1. PRISMA flowchart illustrating the screening and selection of papers for analysis.

3.7. Assessment of reporting bias

Funnel plot could not be carried out since there were insufficient studies in the review despite a comprehensive search strategy. The search strategy was followed with a broad inclusion of all types of study designs in order to unearth all the available studies pertaining to our research question.

4. Results

4.1. Study characteristics

Nine studies (Table 1) met the inclusion criteria with a total of 2045 cases, of which 1369 patients were included in the final analysis after removal of non-blushers and those who were not followed up or responded to questionnaires. Sympathetic blockade was achieved by clipping the second sympathetic ganglion above and below, whereas sympathectomy interrupted the ganglia varying from T1 to T3 except in one study where the interruption was from T1 to T5. Two studies compared the outcomes of sympathectomy at T2 and T2–3 [17,23]. In one study [4], 2nd ganglion interruption was performed in pure blushers while the 2nd and 3rd ganglia were interrupted in those with facial blushing with palmar hyperhidrosis. In this study, split data for facial blushing and hyperhidrosis was available. Follow-up period was 1–60 months with a median of 20 months in one study [17] and mean of 21 months in the remaining 8 studies. Eight of the nine studies reported some formal method of preoperative assessment of facial blushing. All the nine studies used patient opinion, visual analog scores and/or a standardized questionnaire postoperatively to report outcome of surgery. Eight studies performed sympathectomy and one sympathetic blockade by clipping. All studies but one documented the procedure to be bilateral.

The two intervention trials[4,23] and the seven case series [15–21] were analyzed separately (Table 2). Interventional trials showed a higher rate of occurrence of compensatory sweating, recurrence of blushing, relief of blushing, and proportion of patients satisfied with the results from the procedure compared to the case series. Though most of the outcome measures had a lower occurrence in case series compared to intervention studies, few of the outcomes namely, gustatory sweating, Horner's syndrome, proportion of people partially satisfied were higher. Our focus is to analyze outcomes of sympathectomy as mentioned earlier in data analysis. After careful methodological consideration, we decided to combine all the studies for detailed discussion as there is no clear justification for separating the trials.

4.2. Primary outcome measures (relief of facial blushing and patient satisfaction)

A self-report measure which rates the relief of facial blushing (Table 3) was reported as good, partial, and no relief in five

Table 1. Characteristics of studies of the thoracic sympathetic chain interruption as treatment for facial blushing.

Authors	Country, period of study	Type of study	Facial blushing (N)	Age	Diagnostic tests	Surgical procedure
Jadresic E, 2011 [4]	Chile 2003–2009	Prospective, observational, open-label clinical study.	101	13–60	BSPS, SPIN, LSAS, SADS, Blushing severity score	Sympathectomy
Neumayer C, 2005 [15]	Austria 2001–2005	Prospective case series	26	Avg. 35.1	Standardized questionnaire, physical examination and personal interviews.	Sympathetic clipping
Lardinoisa D, 2002 [16]	Switzerland 1996–1998	Prospective case series	18	18 – 67 (Avg. 34)	Visual analog score	Sympathectomy
Licht PB, 2006 [17]	Denmark 1999–2004	Retrospective review	173	16–67 (Med 37)	Questionnaire	Sympathectomy
Drott C, 2002 [18]	Sweden Period Not mentioned	Retrospective study	831	15 – 74 (Avg. 34)	Visual analog scale	Sympathectomy
Fibla JJ, 2008 [19]	Spain 2000–2006	Prospective Case series	24	18–56 (Avg 33.7)	Clinical assessment and questionnaire	Sympathectomy
Saetre T, 2005 [20]	Norway 1992–2002	Prospective case series	44	17 – 53 (Avg. 32)	No data	Sympathectomy
Adair A, 2005 [21]	UK 1998–2001	Retrospective case review	59	8–52	Clinical assessment and visual analog score	Sympathectomy
Licht PB, 2012 [23]	Denmark 2005–2011	RCT	93	18–56 (Avg. 29)	Patient history QOL by SF-36	Sympathectomy

Table 2. Subgroup analysis of intervention trials and case series for outcomes.

Outcome (pooled proportion)	Study design	No of studies	Total sample size	Proportion(95% CI)	I^2
Compensatory Sweating (74.18%)	Intervention trial	2	194	95.77% (81.99%, 100.00%)	0%
	Case series	7	1175	67.66% (47.76%, 87.56%)	89.0%
Gustatory Sweating (24.42%)	Intervention trial	2	194	22.23% (0%,51.33%)	94.5%
	Case series	6	1151	26.20% (8.21%, 44.18%)	94.9%
Dry Hands (61.85%)	Intervention trial	1	93	65.59% (49.13%, 82.05%)	-
	Case series	2	217	61.57% (0%, 134.02%)	95.6%
Horner's Syndrome (1.82%)	Intervention trial	2	194	1.41% (0%, 3.08%)	0%
	Case series	3	101	4.08% (0.14%, 8.01%)	0%
Regretting Surgery (6.79%)	Intervention trial	2	194	6.90% (0%, 17.55%)	86.7%
	Case series	3	235	7.78% (3.78%, 11.78%)	13.4%
Satisfied (84.02%)	Intervention trial	1	101	89.11% (70.70%, 100.00%)	-
	Case series	4	112	79.90% (63.35%, 96.46%)	0%
Partially Satisfied (7.23%)	Intervention trial	1	101	5.94% (1.19%, 10.69%)	-
	Case series	3	94	9.43% (3.22%, 15.63%)	0%
Unsatisfied (6.83%)	Intervention trial	1	101	4.95% (0.61%, 9.29%)	-
	Case series	4	112	10.37% (4.40%, 16.33%)	0%
Relief of Blushing (78.3%)	Intervention trial	1	93	84.95% (66.21%, 100.00%)	-
	Case series	7	1175	77.46% (54.23%, 100.00%)	91.8%
Partial relief of blushing (15.49%)	Intervention trial	1	93	11.83% (4.84%, 18.82%)	-
	Case series	4	302	17.36% (4.83%, 29.89%)	87.6%
No relief of blushing (8.01%)	Intervention trial	2	194	9.05% (0%, 21.38%)	88.0%
	Case series	6	1149	8.05% (4.14% 11.96%)	61.8%
Recurrence of blushing (7.18%)	Intervention trial	2	194	15.01% (0%, 43.52%)	91.6%
	Case series	3	235	3.43% (0.11%, 6.76%)	14.5%

studies. Three studies reported it as relief and no relief and one reported only patients with no relief. Meta-analysis was carried out for the primary outcome measures. The overall pooled estimate of proportion of patients who got relief of facial blushing was 78.30% (95% CI, 58.20–98.39%). There was significant heterogeneity in the results (I^2 = 90.4) and hence we used the random effect model for meta-analysis. Five studies reported the data on partial relief of blushing. The overall pooled estimate of proportion for partial relief was 15.49% (95% CI, 6.32–24.66%). The significant heterogeneity in the results (I^2 = 83.5) justified the random effect model for meta-analysis. Data regarding patients who did not have relief after surgery was pooled and the proportion was 8.01% (95% CI, 4.65–11.38%). Another primary outcome considered in the review is the satisfaction of patients with regard to the surgical procedure (Table 3). The proportion of patients completely satisfied in five studies was 84.02% (95% CI, 71.71–96.33%). In four studies, the proportion of partially satisfied patients was 7.23% (95% CI, 3.46–11.0%). The pooled estimate for unsatisfied patients in five studies was 6.83% (95% CI, 3.32–10.33%). The pooled estimates for relief of facial blushing and patient satisfaction are depicted in the forest plots in Figure 2 (a–d).

4.3. Secondary outcome measures (recurrence of blushing, regretting surgery and complications)

Table 4 shows the meta-analysis of pooled estimates of proportion for the secondary outcome measures (recurrence of blushing, regretting undergoing surgical procedure, compensatory sweating, gustatory sweating, Horner's syndrome, and dry hands). Forest plot of compensatory sweating is depicted in Figure 3.

Table 3. Primary and secondary outcomes.

Citation	No: of patients analyzed	Compensatory sweatingn (%)	Gustatory sweatingn (%)	Dry handsn (%)	Horner's syndromen (%)	Regretting surgeryn (%)	Satisfied n (%)	Partially satisfiedn (%)	Unsatisfied n (%)	Relief of blushingn (%)	Partial relief of blushingn (%)	No relief of blushingn (%)	Recurrence of blushingn (%)
Licht, PB 2012 [23]	93	86(93%)	35(38%)	61(66%)	1 (0.93%)	12 (13%)	n/a	n/a	n/a	79(85%)	11 (11.8%)	3 (3.2%)	28 (30%)
Jadresic E, 2011 [4]	101	100 (99%)	8(8%)	n/a	2(2%)	2(2%)	90(89%)	6(6%)	5(5%)	n/a	n/a	16 (16%)	1(1%)
Fibla JJ,2008 [19]	24	8(33%)	n/a	n/a	1(4%)	n/a	18 (75%)	2(8.3%)	4(16.6%)	20 (83.3%)	0 (0%)	4 (16.7%)	n/a
Licht PB, 2006 [17]	173	153 (88%)	52(30%)	45(26%)	n/a	18(10%)	n/a	n/a	n/a	128 (74%)	28 (16%)	17(10%)	4 (2.3%)
Saetre T, 2005 [20]	44	38(86%)	10 (22.7%)	44 (100%)	n/a	2 (4.5%)	35(79.5%)	4 (9%)	5 (11.3%)	39 (88.6%)	4(9%)	1(2.3%)	3(6.8%)
Neumayer C, 2005 [15]	26	6 (23.1%)	14 (53.8%)	n/a	n/a	n/a	20 (76.9%)	3 (11.5%)	3 (11.5)	25(96.2%)	1(3.8%)	0 (0%)	n/a
Adair A, 2005 [21]	59	54(91%)	19(32%)	n/a	2 (3.4%)	n/a	n/a	n/a	n/a	17(29%)	30(50.8%)	12 (20.33%)	n/a
Lardinoisa D, 2002 [16]	18	12(67%)	5 (30%)	n/a	2 (11%)	1 (5.5%)	17(94.5%)	0 (0%)	1 (5.5%)	16 (88.9%)	0 (0%)	2 (11%)	2(11%)
Drott C, 2002 [18]	831	690(83%)	3 (0.36%)	n/a	n/a	n/a	n/a	n/a	n/a	781 (94%)	0 (0)%	50 (6%)	n/a
Total	1369	Median 86% Range 23.1–99%	Median 30% Range 0.36–53.8%	Median 66% Range – 26–100%	Median 3.4% Range – 0.93–11%	Median 5.5% Range – 2–13%	Median 79.5% Range – 75–94.5%	Median 8.3% Range – 0–11.5%	Median 11.3% Range – 5–16.6%	Median 86.8% Range – 29–96.2%	Median 6.4% Range – 0–50.8%	Median 10% Range – 0–20.33%	Median 6.8% Range – 1–30%

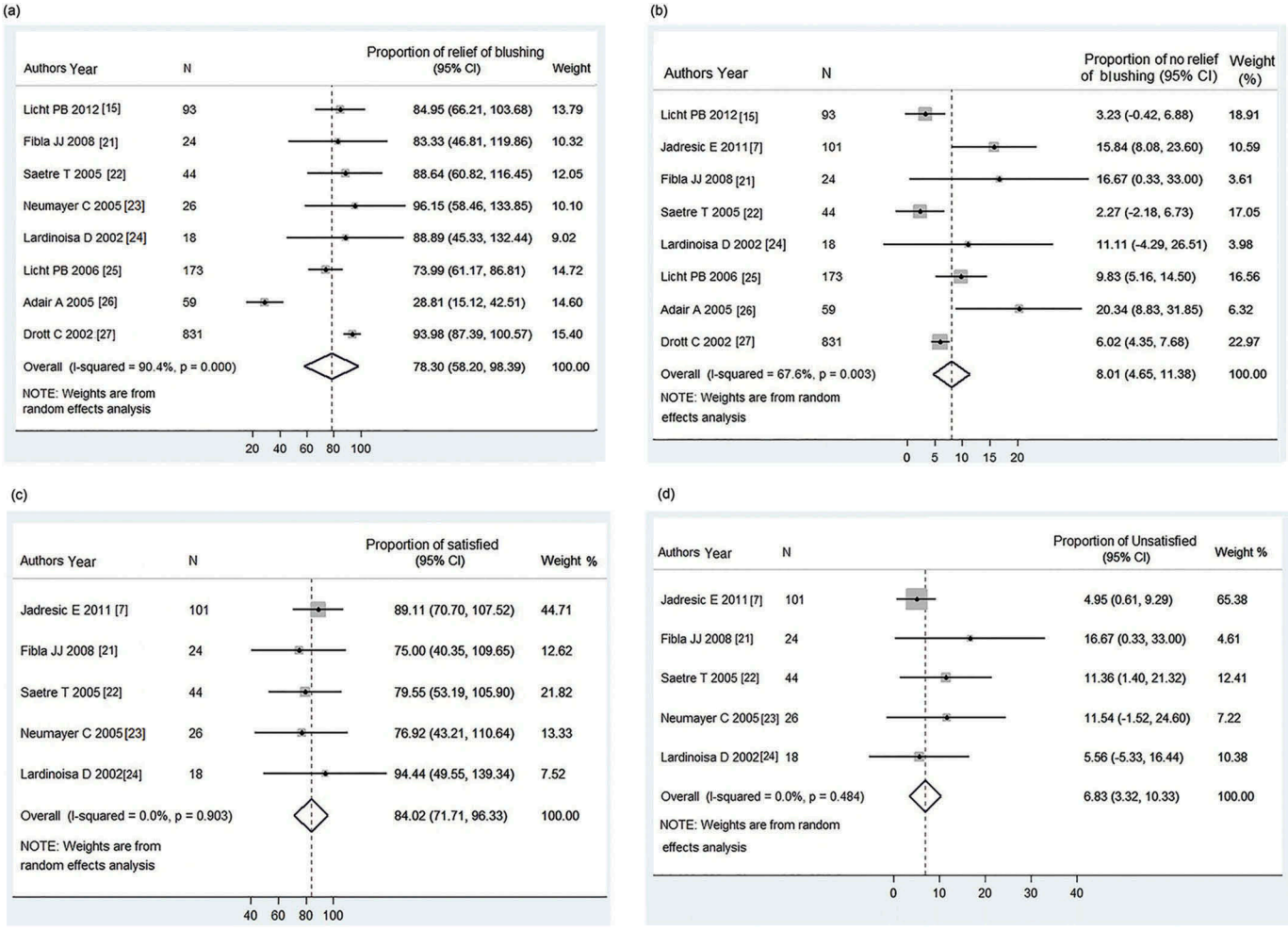

Figure 2. (a) Relief of blushing after sympathetic interruption. (b) No relief of blushing after sympathetic interruption. (c) Satisfied patients after sympathetic interruption. (d) Unsatisfied patients after sympathetic interruption.

Table 4. Results of pooled estimates of proportion, confidence interval, and I^2 of secondary outcomes.

Outcome	No of studies	Total sample size	P(95% CI)	I^2
Recurrence of blushing	5	429	7.18%(1.86%, 12.49%)	85.8%
Regretting Surgery	5	429	6.79%(2.08%, 11.50%)	71.5%
Compensatory Sweating	9	1369	74.18%(58.10%, 90.26%)	88.9%
Gustatory Sweating	8	1345	24.42%(12.22%, 36.61%)	94.9%
Horner's Syndrome	5	295	1.82%(0.28%, 3.36%)	0%
Dry Hands	3	310	61.85%(21.89%, 101.81%)	94.6%

There were no deaths reported in any of the series. The other uncommon complications reported were port site pain, chest pain, non-gustatory facial sweating, pneumothorax, intraoperative bleeding and pulmonary embolism with paradoxical embolism causing stroke, impaired physical performance, palpitations, arm pain, chills, dizziness, GI disturbances, and headache[15–18,21].

5. Discussion

Facial blushing is a socially embarrassing problem, especially in fair-skinned individuals. It is involuntary and occurs in response to varying degree of emotional stimuli which can lead to a severe social impact on the affected individual [5]. Although often used interchangeably with facial flushing, the latter is a secondary response to specific causes such as dermatological conditions, heat, alcohol, sepsis, and autonomic nerve injuries as occurs in Harlequin syndrome [26,27] and Frey's syndrome[28] (Syn: Baillarger's syndrome, Dupuy's syndrome, Frey-Baillarger's syndrome or auriculo-temporal syndrome). Flushing is devoid of a psychological component.

Nonsurgical treatment in the form of psychotherapy such as cognitive therapy, task concentration training [9,29,30], psycho-educational group intervention [31], and cognitive behavioral group therapy [32], or pharmacotherapy with drugs such as sertraline or beta blockers [4], Botulinum toxin A [33] are often ineffective and patients then seek surgical intervention.

Endoscopic transection or clipping of the sympathetic chain at T2 or T2–3 level is increasingly being offered to patients with facial blushing and/or facial sweating. NICE guidelines recommend endoscopic sympathectomy as

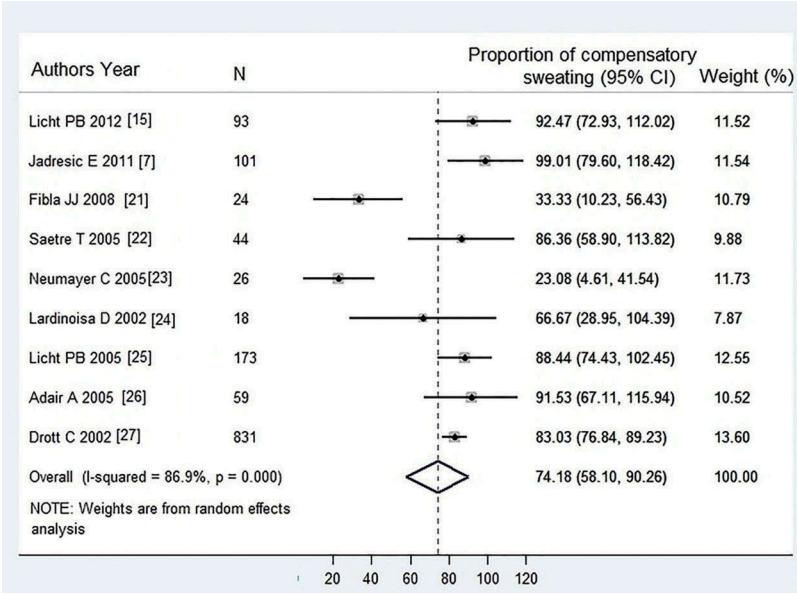

Figure 3. Occurrence of compensatory sweating after sympathetic interruption.

treatment of patients with facial blushing [34]. The evidence to this effect that we retrieved was mostly in the form of case series.

The success rate of sympathectomy or sympathetic interruption, described as relief of blushing in various studies systematically reviewed was 78.30%. Occasionally, relief of symptom is achieved on one side of the face whereas the other side remains blushed requiring re-operation [15]. Another significant primary outcome was the complete satisfaction reported by 84.02% of patients who underwent sympathectomy. Comparison of the success rate of sympathectomy (78.30%) to patient satisfaction rate (84.02%) probably suggests a placebo effect. Compensatory sweating was reported in most studies (74.18%, C.I 58.10–90.26%) but only 6.79% of patients regretted having had the procedure. Jadresic et al[4] and Fibla et al. [19] have reported 13% and 8% of severe compensatory sweating, respectively, after sympathectomy for facial blushing. This would reflect that thoracic sympathetic interruption is an effective procedure for facial blushing.

Gustatory sweating was the next commonest side effect reported, but the incidence is much lower than compensatory sweating (24.42%). Horner's syndrome was reported in just under 2% of cases. This was thought to be due to thermal injury of stellate ganglion during transection of the chain by diathermy. This may probably be reduced by using ultrasonic scalpel [23] or cutting diathermy instead of coagulation [16] and by clipping [4]. In a study by Chou et al. [35], which was not included in the analysis due to poor quality, 16.7% (9 patients) regretted the procedure and underwent reversal by removal of clips with variable improvement in compensatory sweating. To facilitate reversal of the procedure, recent studies have promoted sympathetic clipping, but there is no clear evidence to support this [35–37]. An experimental study by Loscertales et al. [38] on an animal model concluded that the absence of nerve regeneration suggests that clipping of nerve is not a reversible procedure. Cardiovascular complications such as bradycardia, palpitations, and hypotension have been reported after thoracic sympathectomy occasionally and is usually asymptomatic. In a subgroup of 18 patients who underwent T2–4 sympathectomy for palmar hyperhidrosis, Drot et al. [39] reported a significantly reduced heart rate and systolic blood pressure. This had no effect on maximal workload. In our review, the only cardiovascular complication in the form of palpitations was reported by Drot et al. [18] in 0.7% of their patients.

The only RCT in this review [23] suggests there is no difference between interruption at the second or the second and third ganglia with respect to complications. Whereas, a retrospective questionnaire-based study by the same authors in an earlier publication suggested T2 sympathectomy to be 'superior' to T2–3 sympathectomy because of lesser side effects [17]. The RCT would obviously carry more significance. The society of thoracic surgeons expert consensus for the surgical treatment of hyperhidrosis [14] suggests interruption at the level of the third ganglion for facial hyperhidrosis and we assume that this would be the case for blushers as well. Higher the sympathetic interruption, greater is the risk of compensatory sweating [35]. The new Lin-Telaranta classification [40] suggests sympathetic interruption at the second ganglion for facial blushing.

No deaths were reported in the selected studies in our review. Ojimba [41] however reports mortality secondary to massive intra-thoracic hemorrhage or anesthetic complication following endoscopic thoracic sympathectomy. Ojimba also reports residual pneumothorax requiring chest drain (0.4–2.4%) and surgical emphysema (2.7%) [41]. In our review, we found dry hands in 2 studies 61% and 26% [17,23], port site pain in 2 studies 2% and 3.7% [15,17], postoperative chest pain in 2 studies, 88% and 0.48% [18,21], non-gustatory facial sweating in one occurring in 11.8% of patients [21] and pneumothorax, intraoperative bleeding and pulmonary embolism with paradoxical embolism causing stroke occurring in one out of 18 patients (5%) in one study [16]. One study [18] also noted impaired physical performance (1.3%), palpitations (0.7%), arm pain (0.6%), chills (0.6%), dizziness (0.36%), GI disturbances 0.36%,

and headache 0.12%. Two studies [17,21] reported chest drain insertions in 1% (1/95) and 8.5% (5/589) patients, respectively.

In view of invasiveness of endoscopic thoracic sympathectomy, its potential complications and postoperative dissatisfaction, case selection is important. There is a compelling need to define strict criteria in the diagnosis of these patients and select appropriate cases for surgical treatment [42–44]. Facial blushing can be self-perceived (fear of blushing) or observed [29] which need to be distinguished before invasive treatment is offered [42]. In cases of perceived facial blushing, various methods of psychotherapy are often effective [9,30–32]. High scoring in social anxiety scoring systems may help identify patient who would benefit from psychotherapy support [45]. Selection of actual facial blushers for surgical treatment can be difficult as patient reported blushing has not matched the objective facial blood flow [5]. Criteria to establish a firm diagnosis and when to offer surgical treatment should be a topic for future research.

5.1. Limitations of this review

The review consisted of 4 good quality and 3 fair case series. The only RCT in this review was of level IIA and one nonrandomized study was of level IIB evidence. The case series were of Level IV evidence. The paucity of good-quality studies could be seen as a limitation to making rigid conclusions.

6. Conclusion

Sympathetic interruption appears to be an effective treatment for facial blushing. However, the treatment should be undertaken after careful scrutiny of patients and indications for treatment. Patients should be explained the occurrence of complications, especially, compensatory sweating and gustatory sweating and the potential failure to achieve any symptom relief. There is a need for better quality evidence to make firm recommendations.

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Declaration of interest

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